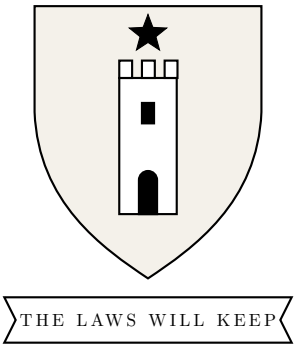
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ON DECIDABLE MATTERS, WITH AN ANSWER TO SEVERIN'S FIFTH QUESTION

By Cummings

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ON DECIDABLE MATTERS, WITH AN ANSWER TO SEVERIN'S FIFTH QUESTION

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Received in the Month of Reawakening, F.E. 318, the Year of Dense Fog.

Summary. The reckonable numbers are defined as the numbers whose figures can be written out, one after another, by a rune chain, and reasons are given for holding that every number which can be calculated by rule is of this kind. Three matters are then shown to be undecidable: whether an arbitrary rune chain is inexhaustible; whether it will come to a stop on an arbitrary input; and whether it will ever write a given mark. In the sense here given to the word "means", the answer to Severin's Fifth Question is negative. It follows further that no non-trivial property of the behaviour of chains admits a total judge; that chains confined to a fixed number of cells can be judged in principle; and that every system whose proofs can be listed by rule, and whose assertions that a chain never stops are sound, leaves some such assertion unproved. Section 12 treats of insight. An appendix shows that "reckonable" in the present sense and "recursively definable" in the sense of Prescott describe the same functions.

Introduction

The "reckonable" numbers may be described, shortly, as the numbers whose figures can be worked out one after another by finite means. Though this is a paper on decidable matters, it begins with reckonable numbers, for the two are handled by one method. It is equally easy to define reckonable functions of an integral variable, and decidable classes of matters, each member of which can be settled by finite means as holding or not holding. By my definition, a number is reckonable if some rune chain will write down its figures one by one.

In §9 I give reasons for supposing that every number which can be calculated by rule is reckonable, and in §10 I show that the numbers actually met with in star-tables, tide-tables and arithmetic — the real roots of algebraic equations, the values of the smooth counting formula, the zeros of the density function — are all reckonable. Numerous as they are, the reckonable numbers can be numbered off one by one, and so do not include all real numbers; in §8 a definite number is exhibited which is not reckonable.

Finally I apply these results to Severin's Fifth Question, whose words are: "Is there a means by which, before any formula is cast, its outcome may be known?" [1]. When the question was put, a formula meant a spell or an array, and its outcome meant whether it would come to rest or go on working for ever — whether, for instance, a self-adjusting ward would adjust itself without end. The difficulty lies in the word "means". To show that there is a means one need only produce it; to show that there is none, one must first say what sort of thing a means is.

*Kanaich Institute of Technology, Kanaich. Section 7 is printed as corrected by the author's note in vol. 224, part 1.

About F.E. 200 Ashby laid down the universal rune units and the three rules by which they are joined: concatenation, branching and looping [2]. Since then every engine formula can be written as a rune chain in universal units, and every engine type which realises all the units can carry out any rune chain whatever, given enough cells and time. The question may therefore be asked more narrowly: is there a rune chain which, given any rune chain and an input to it, will decide whether the latter will ever come to a stop? Such a chain I shall call a total judge.

The question is not an idle one. Since the universal units became the industrial standard for formulae, an error in the common library of formulae spreads to every city; with a total judge, each formula could be judged before it entered the library. Finley's proofs of safety for finite-state control formulae in 213 [3], and Morton's self-checking chains of 288 [4], were steps in that direction. A total judge would also turn many hard questions into a single judgment. Make a chain which, starting from four, tests each even number in turn to see whether it is the sum of two primes, and stops at the first that is not: whether it stops is precisely Severin's Second Question. Nolan's proof that it never stops cost him three years merely to mend a gap in the first draft, and the revised draft was examined for ten years more; with a total judge one judgment would have sufficed. Aldin's conjecture, too, can be turned into the question whether a certain chain stops. A total judge would be too good to be true, and I shall show that it does not exist. Severin himself said that to learn which roads lead nowhere is part of the proper business of learning.

Prescott, of White Tower University, has lately defined by quite different means the "recursively definable functions" [8]. The two definitions differ in appearance but pick out the same functions; the proof is given in the Appendix.

Sections 1–5 define rune chains, reckonable numbers and chain numbers, and give two examples of chains; §§6–7 describe the universal chain; §8 uses the diagonal method to prove three matters undecidable; §§9–10 discuss the extent of what can be done by rule; §11 answers the Fifth Question and draws some consequences; §12 treats of insight.

1. Rune chains

A rune chain works upon a storage band divided into cells, each of which can hold one mark taken from a finite table of marks; the band has a beginning, and may be extended to the right as required, without limit. At each step the chain is reading one cell of the band, which I call the cell under reading, and the mark in it, the mark under reading; at the same time the chain is in one of its own units, which I call the present unit. What it does next is fixed by the present unit and the mark under reading alone: it may write a mark in the cell under reading, or erase the mark there; it may move one cell to the left or to the right, or not at all, a move to the left from the first cell leaving it where it is; and it then passes into some unit.

Ashby's universal units [2] are units of exactly this kind: each prescribes, for each mark that may be read, what is to be written, which way to move, and which unit to pass into. Concatenation is the passage from one unit to the next when it has done its work; branching, the passage to different units according to the mark read;

looping, the return to a unit already passed through. Every rune chain written in universal units may therefore be regarded as a chain in the present sense. What is taken here is only the part of the universal units that does the reckoning; where a unit is joined to a mana engine or an array — its supply, its warrant and its coupling — is treated in this paper only as a command the chain writes down, however it may be carried out in the engine.

A chain has a start, the unit in which it begins, and a stop. To pass into the stop is to come to a stop, and the contents of the band are then the output. A chain may also never reach the stop, going round some loops for ever; it is then said never to stop.

Actual calculating engines have a finite number of cells: when these are used up the engine either halts or asks to be furnished with more. I consider abstract chains whose band may be extended at need, because my subject is everything that can be done by rule, and it would be wrong to fix in advance how many cells such doings may use. The case of a fixed number of cells is treated in §11, Theorem 5.

2. Definitions

Self-moving and awaiting chains. If at each step a chain's next move is fixed by the present unit and the mark under reading alone, I call it self-moving. Some engines of old needed, at certain branchings, a command from the keeper of the engine before they could tell which unit to pass into; this was called awaiting command, and such chains I call awaiting chains. Only self-moving chains are considered here, and "chain" will mean a self-moving chain.

Figure-writing chains. Some chains write two kinds of marks upon the band: figures, which are 0 and 1 only, and scratch marks, which the chain uses for its own working. Figures are written in the odd cells and, once written, are never altered; scratch marks are written in the even cells and may be written and erased at will. Figures must be written in the odd cells in order from the first, from left to right, none being passed over; a chain that passes one over also fails to keep the rules of figure-writing.

Inexhaustible and exhaustible chains. A figure-writing chain started on a blank band is inexhaustible if it writes infinitely many figures; if it writes only finitely many, whether it stops or not, it is exhaustible. A chain that does not keep the rules of figure-writing — that alters or erases a figure once written, for instance — I count as exhaustible, so that "inexhaustible" is defined for every chain number.

Reckonable sequences and numbers. The figures written by an inexhaustible chain, taken in order, form an infinite sequence of 0s and 1s, which I call the sequence reckoned by the chain. Any sequence reckoned by some inexhaustible chain is a reckonable sequence. If "0." is set before a reckonable sequence and the whole is read as a binary fraction, the number obtained is a reckonable number; an integer added to a reckonable number is also called a reckonable number.

Reckonable functions. A function f is reckonable if there is a chain which, given the written form of any integer n, always comes to a stop with the written form of $f(n)$ on its band.

Decidable matters. A class of matters is decidable if each matter of the class can be written as a string of marks, and there is a chain which, given the written form of any one of them, always stops, writing "yes" if the matter holds and "no" if it does not.

3. Examples of chains

Of the two chains below one is inexhaustible and the other comes to a stop. Each is written as a table, every row of which gives a present unit and a mark under reading, the action then taken, and the next unit; "blank" means that the cell holds no mark.

Example 1. A chain writing 0 1 0 1 …. It has four units and starts in a. Figures are written in the odd cells; the even cells are left blank, since this chain needs no scratch marks.

TABLE 1. Example 1.

Present unit	*Mark read*	*Action*	*Next unit*
a	blank	write 0, move right	b
b	blank	move right	c
c	blank	write 1, move right	d
d	blank	move right	a

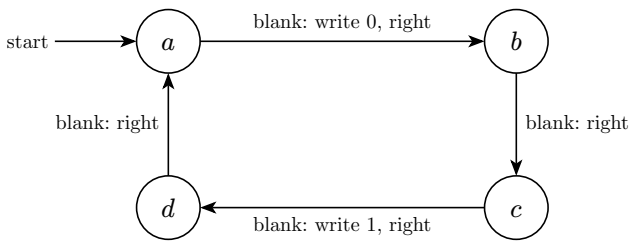


FIG. 1. The four units of Example 1. Each arrow is labelled with the mark read and the action taken.

This chain never stops and writes infinitely many figures; it is inexhaustible. The sequence it reckons is 0 1 0 1 …, and the number it reckons, read as a binary fraction, is $0.0101\ldots = 1/3$. A star-reckoning array unfolding a star-table by a fixed rule does work of just this kind.

Example 2. Addition in single strokes. The band bears a string of strokes, a plus sign and a second string of strokes, for instance "111+11", and the chain begins at the leftmost cell. It turns the plus sign into a stroke, runs to the right-hand end, erases the last stroke, and stops; the number of strokes left is the sum of the two numbers.

Table 2. Example 2.

Present unit	*Mark read*	*Action*	*Next unit*
a	1	move right	a
a	+	write 1, move right	b
b	1	move right	b
b	blank	move left	c
c	1	erase	stop

When the second number is nought, as in "111+", unit b reads a blank at once, moves left on to the stroke that was the plus sign, and erases it, which is again right. Cases not listed, such as unit c reading a blank, never arise with a proper input; for completeness they are all to be understood as passing into the stop.

4. Sub-chains

When a chain grows long its table becomes hard to read. Steps much used may be written as sub-chains with parameters, like functions in a formula: when the parameters are given, a definite table results. Four are used below. They all work within a bounded stretch of the band: a chain that uses them first writes a mark of beginning in the first cell, and an end mark after the last cell it has written, moving the end mark to the right whenever the band is extended.

⟨Seek a, b, m⟩. From the cell under reading go left to the mark of beginning, then right, cell by cell, to the first cell marked m; if one is found, stop there and pass into a; if the end mark is reached without finding one, pass into b.

⟨Erase a, b, m⟩. Find, as with ⟨Seek⟩, the first m, erase it and pass into a; if there is none, pass into b.

⟨Copy a, b, m, n⟩. Find the stretch beside which m is marked, and copy its marks cell by cell into the blank stretch marked n, keeping the place with a further scratch mark that is erased when the copying is done; then pass into a, leaving m where it was. If there is none, pass into b. A stretch of marks ends at a mark of division.

⟨Compare a, b, c, m, n⟩. Compare the two stretches of marks beside which m and n are marked; if they agree pass into a, if not into b, and if either m or n is not found, into c. Each stretch ends at a mark of division.

Each sub-chain uses only finitely many units and, when expanded, is still a finite table. To use a sub-chain within a chain is to copy its table in, putting actual units and marks for the parameters. The universal chain of §7 is written with them.

5. Chain numbers

The table of a chain can be put into a standard form. Let the units be called $u_1, u_2, u_3, ...$, the start being u_1 and the stop being marked separately; let the marks be called $m_0, m_1, m_2, ...$, m_0 being the blank, m_1 the figure 0 and m_2 the figure 1, each mark keeping the same number throughout, so that if a chain does not use some mark no other mark moves up to take its number; and let each row of the table be written as five items — present unit, mark read, mark written, move (left, right or none), next unit — the rows following one another with a separating

sign between them. Every chain then has a standard form, written with only a few letters.

Write the standard form in the marks of the Great Library's Code Table [7]. Every mark in the Code Table has a code of the same number of figures, short codes being filled out with noughts in front; write the codes of the marks one after another and put a 1 in front of the whole. The number obtained I call the chain number of the chain. Because the codes are all of one length, a chain number can be cut back, piece by piece, into the marks it came from, so that different standard forms have different chain numbers. A pair of numbers (c, x), a pairing sign being set between the two, is made into a single number in the same way, to serve as an input. Not every number is a chain number, but whether a number is one can be tested by a chain; a number which is not a chain number shall be taken to stand for a chain that stops at once. All chains can therefore be arranged in a single list in the order of their chain numbers: chain 1, chain 2, chain 3, ….

It follows that the reckonable sequences can be numbered off as well, since every reckonable sequence is reckoned by at least one inexhaustible chain and the chains can be numbered off. The real numbers cannot be so numbered, as classical arithmetic proved long ago. There are therefore numbers which are not reckonable, and in §8 I shall exhibit a definite one by the same diagonal method.

6. The universal chain

LEMMA 1. *There is a chain, ⟨Universal⟩, such that for any chain number c and any input x: if chain c stops on x, ⟨Universal⟩ stops on (c, x) and writes the output of chain c (the marks of chain c being written in their standard form of § 5, and the figures 0 and 1 as they are); if chain c never stops on x, neither does ⟨Universal⟩ on (c, x).*

⟨Universal⟩ need not be made afresh for each chain: it reads the chain number and follows, step by step, the table that the number spells out. This is just what Ashby's transcription tables do. A transcription table realises every universal unit upon one engine type; ⟨Universal⟩ takes the rune chain itself for its engine type and realises every rune chain. An engine type which can carry out ⟨Universal⟩ can carry out every rune chain, given enough cells and time. The details of the construction follow.

7. Details of the universal chain

⟨Universal⟩ divides its band into three regions (Fig. 2). On the left is the standard form of chain c, spelled out from its chain number; this is the text region. Next comes the marker region, which records the unit chain c is in. On the right is the simulation region, which holds chain c's own band. The table of marks of ⟨Universal⟩ is fixed and finite; the units and marks of chain c, in whatever region, are written in the standard form of § 5. In the simulation region every mark of chain c is filled out to one length, that of the longest mark of chain c, with noughts in front; so that one cell of chain c is, on the band of ⟨Universal⟩, a stretch of fixed length, which I call a simulated cell. The simulated cell chain c is reading has the scratch mark R

beside its first cell on the right. R, like the mark L below, is a scratch mark and is written in an even cell. The regions are parted by marks of division, and the marks of beginning and end are as in § 4. ⟨Universal⟩ keeps the arrangement of figure-writing chains: the codes of all three regions are written in the odd cells, and the even cells hold only scratch marks such as R and L, so that R, set beside the first cell of a simulated cell, covers no code. The units and marks of chain c in the text and marker regions are filled out to the same length as in the simulation region, so that the comparison of step (ii) can be made.

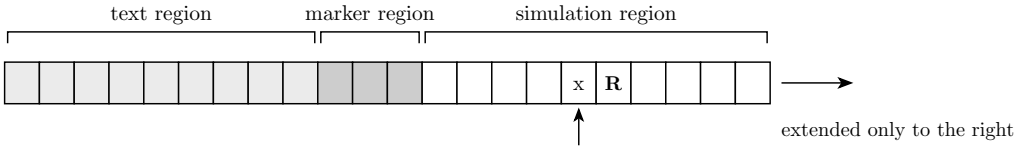


Fig. 2. The band of ⟨Universal⟩: the text region on the left, the marker region in the middle, the simulation region on the right. The simulation region is extended only to the right.

At the outset ⟨Universal⟩ tests whether the c of its input is a chain number, stopping at once if it is not; if it is, it spells c out into its standard form as the text region, finds the length of the longest mark of chain c, writes the start u_1 in the marker region, writes x into the simulation region in simulated cells, and marks the first of them with R.

⟨Universal⟩ does one thing over and over, each round in four steps.

(i) By ⟨Copy⟩, copy the mark in the simulated cell marked R, cell by cell, into the marker region, after the present unit, so as to form the pair "present unit, mark read"; R stays where it was.

(ii) By ⟨Compare⟩, compare this pair with the first two items of each row of the text region in turn, and mark with L the row that agrees. The text is a complete table, so such a row is always found; should none be found, the convention of §3 applies and the stop is entered.

(iii) Following the last three items of the row marked L, write the new mark in the simulated cell marked R, and move R to the simulated cell on the left or on the right; if the move is none, R does not move, nor does it when the move is to the left from the leftmost simulated cell. When R is moved beyond the rightmost simulated cell, a blank simulated cell is added at the right, and the end mark moves with it.

(iv) Rewrite the marker region as the next unit of that row, and erase L. If the next unit is the stop, first erase the text and marker regions, strip the simulated cells of the noughts that filled them out, move the contents of the simulation region, in the standard form of § 5, to the beginning of the band, and then stop.

For each step of chain c, ⟨Universal⟩ performs one round of these four steps, each round being finished in finitely many steps; and when it is finished the simulation region still holds the band of chain c, R still marks the simulated cell chain c is reading, and the marker region still records the unit chain c is in. If chain c stops

on x, therefore, ⟨Universal⟩ stops and writes the output of chain c; and if chain c never stops, neither does ⟨Universal⟩. This proves Lemma 1.

8. Application of the diagonal method

THEOREM 1. *There is no chain which, given any chain number, will decide whether that chain is inexhaustible.*

Proof. Suppose there were such a chain, ⟨Discern⟩, which given any chain number always stops, writing "yes" if the chain is inexhaustible and "no" otherwise. With it construct a figure-writing chain ⟨Diagonal⟩ which works in rounds, the n-th round being as follows. By ⟨Discern⟩, test the chains in order from chain 1 until the n-th inexhaustible chain is found; by ⟨Universal⟩, run that chain step by step, counting after each step the figures it has written, and break off when it has written n; and write on the band the opposite figure, 1 if it was 0 and 0 if it was 1. The workings of ⟨Discern⟩ and ⟨Universal⟩ are carried on by ⟨Diagonal⟩ in answering scratch marks in the even cells; in the odd cells it writes only the figure reversed in each round, which is never altered.

Every round is completed in finitely many steps, for ⟨Discern⟩ always stops, and the n-th inexhaustible chain, being inexhaustible, always writes its n-th figure. ⟨Diagonal⟩ therefore writes infinitely many figures and is itself inexhaustible, so that it stands somewhere in the list of all inexhaustible chains, say as the k-th. But then its k-th figure is, by its construction, the opposite of the k-th figure written by the k-th inexhaustible chain — that is, by itself — which is absurd. Hence there is no ⟨Discern⟩.

The sequence obtained by reversing, for each n, the n-th figure of the n-th reckonable sequence differs from every reckonable sequence in at least one place, and is therefore not reckonable. With "0." set before it, it gives the definite number, not reckonable, promised in §5. By the n-th reckonable sequence I mean that reckoned by the n-th inexhaustible chain in the order of chain numbers; if one sequence is reckoned by several chains it appears several times, which does no harm. Only a terminating binary fraction has two forms, the second ending in an endless run of 1s; both forms are in the end periodic, and both are reckonable sequences. The sequence obtained above, not being reckonable, is neither form of any terminating fraction; the number it gives therefore has one form only, and is not reckonable.

THEOREM 2. *There is no total judge. By a total judge I mean a chain ⟨Judge⟩ which, for every pair (c, x), stops, writing "stops" if chain c stops on x and "never stops" if it does not.*

Proof. Suppose ⟨Judge⟩ exists. Construct a chain ⟨Contrary⟩ which, given a number c, (i) copies c, with the pairing sign between, so as to have (c, c); (ii) concatenates ⟨Judge⟩ and obtains its output on (c, c), which by hypothesis it surely does; (iii) branches on that output, entering a loop it never leaves on "stops", and passing into the stop on "never stops". ⟨Contrary⟩ is made from ⟨Judge⟩ by the three rules of combination and is therefore a chain; let its chain number be r.

Consider ⟨Contrary⟩ on the input r. If it stops on r, the output seen at step (iii) was "never stops": ⟨Judge⟩ has declared that chain r never stops on r, and chain r is

⟨Contrary⟩, which is a contradiction. If it never stops on r, the output seen at step (iii) was "stops": ⟨Judge⟩ has declared that ⟨Contrary⟩ stops on r, which is again a contradiction. Hence ⟨Judge⟩ does not exist.

The astrologers below are often confounded by a single remark: "You say I shall go out tomorrow; then I shall stay at home." Whatever the astrologer foretells, a hearer who does the opposite will make the prophecy fail. ⟨Contrary⟩ is just such a hearer, who listens to the verdict and does the reverse.

Theorem 2 does not say that whether a chain stops can never be known. If a chain will stop on some input, one has only to let it run, and sooner or later one sees it stop: stopping is something that can be waited for. What cannot be waited for is the other side. If a chain never stops, no length of running will show that it will not stop later; and the whole difficulty of a total judge lies on that side. Theorems 1 and 2 are two faces of one obstacle.

Theorem 3. *There is no chain which, given any chain number, will decide whether that chain, started on a blank band, will ever write a given mark, say "0".*

Proof. For any pair (c, x) construct a chain B which, starting on a blank band, first writes x and then carries out the working of chain c on x. The marks B uses are kept wholly apart from "0": x and every mark of chain c are written by B in scratch marks of their own, and where chain c reads or writes a mark B reads or writes the answering scratch mark, so that in the course of the working B never writes "0"; only if that working stops does B write "0". The table of B is obtained from (c, x) by rule. Then B will at some time write "0" if and only if chain c stops on x. A chain deciding whether "0" is ever written would thus give a total judge, contrary to Theorem 2.

9. The extent of what can be done by rule

I now ask whether everything that can be calculated by rule can be calculated by a rune chain. This cannot be proved, since "by rule" is not a notion of arithmetic; I can only give reasons. They are three: a direct look at what a person does who calculates by rule; a comparison with other exact definitions; and the facts of practice.

(I) The direct argument. Of old the Tower of All Things reckoned its star-tables by the Ring Rite. The presiding Grand Master stood at the station of the chief star, and some scores of wizards at the star stations of the array; each did one fixed step only, taking the mana and the token passed to him by the one before, changing them as prescribed, and passing them on. A reckoning might last for months, yet no wizard needed to understand the whole of it; he needed only to understand his own step.

So it is with a reckoner working alone by rule. Let us consider him point by point.

(a) He uses only finitely many kinds of mark. A mark is written within a cell and is of bounded size, and his eye can distinguish only so finely; were there infinitely many kinds, two of them within these bounds would differ by as little as one pleased, and he could not tell them apart. The marks he can use are therefore finite in number; longer writings may be regarded as strings of these marks.

(b) At any moment he looks at only finitely many cells. However much he can take in at a glance, there is a limit, and beyond it he must look more than once.

(c) The states of his mind relevant to the work are finite in number. Since the rules are fixed in writing, what he keeps in mind is only which step of the rules he has reached, and of such steps there are finitely many. Whatever must be remembered that grows as the work goes on — how far he has counted, where he has got to — is written on the paper, not kept in mind (see (e)).

(d) At each step he alters only finitely many cells, moves his eyes only a little way, and passes to the next step of the rules. Any larger action may be broken into several such steps.

(e) The state of his mind may itself be written down. When a reckoner of the Tower goes off duty he leaves a note saying which step he has reached, and the one who comes on carries on from the note, with no need to know what passed through the other's mind. The state of mind can thus be exchanged for marks on paper: a reckoning may be carried on by many people in turn, each working only from the note and the rules.

Taken together, these show that each step a reckoner takes by rule is to read a few cells, to alter and move according to the rules, and to pass to the next step; and this is just what a chain of §1 does at each step. Everything that can be done by rule can therefore be written as a rune chain.

(II) Agreement with other definitions. The "system whose rules can be listed" of the Babel proof of the Fourth Question [5] is one whose proofs can be listed by a chain: write out strings of marks in order of length, test each to see whether it is a correct derivation by the rules, and output it if it is. This requires that whether a derivation is correct can always be settled in finitely many steps; and the system of the Babel proof is of this kind. Prescott's recursively definable functions [8] are, as the Appendix shows, the same as the reckonable functions. That definitions starting from such different places should meet is evidence that they have caught the same thing.

(III) Practice. Every method of judgment the Alliance has so far used — the cross-checking of star-tables, the reconciling of accounts, Finley's checker [3] — has been written as a rune chain; and I can think of no judgment by rule which could not be broken into fixed steps like those at the star stations.

Insight is not included in this; see §12.

10. Large classes of reckonable numbers

The reckonable numbers include almost every number used in actual calculation. I mention some large classes.

(i) Integers and fractions. The binary expansion of a fraction is in the end periodic, and a chain need only repeat the period.

(ii) Real roots of algebraic equations. Let an equation with integral coefficients first be freed of repeated factors, so that its roots are all simple; the number of real roots between two fractions can be counted by the old method of algebra, and the roots can thus be isolated one by one. A root once isolated is sought on the

lattice of binary fractions. Each binary fraction met with is first substituted: if it gives nought, it is the root, and is written as in (i); if not, the root lies between two neighbouring binary fractions of k places, and its first k figures are fixed. The middle point is then substituted, to see in which half the root lies, and each halving fixes one figure more.

(iii) Limits obtained by the calculus of flux, provided the speed of convergence can be reckoned. Let the steps of an approximation be fractions given uniformly in n by a chain, and let it be possible, for any p, to reckon a step beyond which every step differs from the limit by no more than 2^{-p}. Then enclosures of any accuracy can be found, and the figures fixed as at the end of (v). The value of Crossley's smooth counting formula at any integer is of this kind. So is the solution of Aldridge's tide equations over a stretch of time, provided the initial state and the constants are reckonable, the existence and uniqueness of the solution over that stretch have been proved, and its errors can be bounded by reckoning; whether the solution exists and remains smooth over every finite time is the fourth of the Millennium Seven, which this paper does not touch.

(iv) The zeros of the density function. The density function can be evaluated to any accuracy, with a reckonable bound to the error; and by the counting of zeros round a contour the number of zeros within a small rectangle of the complex plane can be found, and found in finitely many steps whenever no zero lies on its edge; in dividing a rectangle one takes several lines of division very close together near the point of division and reckons them by turns, and since the zeros are isolated, one of these lines has no zero on it and is settled in finitely many steps. Every non-trivial zero can therefore be enclosed in smaller and smaller rectangles, so that its real and imaginary parts are reckonable. Aldin's conjecture says that all these zeros lie on one line. Were there a zero off the line, it would lie within some rectangle that does not touch the line and has no zero on its edge; a chain may count the zeros in all such rectangles by turns, advancing each a little at a time and waiting long on none, and stop when it finds a count that is not nought. The conjecture fails, then, if and only if this chain stops, as was said in the Introduction. On the other hand, no number of zeros found on the line can prove that all lie there.

(v) The sum, difference, product and quotient (the divisor not being nought) of reckonable numbers are reckonable. In writing out the figures one first calculates to sufficient accuracy before fixing each figure, lest a later carry alter figures already written. But if the result lies exactly upon a point of division, an enclosure however narrow may always straddle the point, and that figure cannot be fixed. What is asserted here, then, is asserted of each number singly: if the result is not a binary fraction, a narrow enough enclosure will always avoid the points of division; if it is, its terminating expansion is written as in (i). Whether the result lies exactly upon a point of division no single chain can decide.

The numbers reckoned by star-reckoning arrays and calculating engines are thus all reckonable, and the limits found here put no hindrance in the way of the calculations of these classes. Section 8 gives one way of describing a number that

is not reckonable; such numbers are far more numerous than the reckonable, and need not all be made on purpose.

11. The answer to Severin's Fifth Question, with consequences

The Fifth Question asks about any formula whatever. A rune chain is itself a formula, being exactly the formula that an engine executes; so a means of knowing the outcome of any formula before it is cast would decide, for every rune chain and every input, whether the chain stops. By §9 such a means, if it works by rule, can be written as a rune chain, and would then be a total judge, which by Theorem 2 does not exist. In the sense here given to the word "means", the answer to Severin's Fifth Question is therefore: there is no such means.

Some further consequences follow.

THEOREM 4. *Let P be a property of the behaviour of chains; that is, whether a chain has P depends only on the inputs on which it stops and the outputs it then gives, not on the way it is written. If some chains have P and some do not, there is no chain which decides, for every chain, whether it has P.*

Proof. Suppose there were such a chain ⟨Judge-P⟩. The chain that never stops on any input either has P or has not; suppose it has not (otherwise use "not P" in place of P), and take a chain A that has P. For any pair (c, x), construct a chain B which, given y, first stores y in scratch marks of its own at the beginning of the band and carries out the working of chain c on x to the right of it; and, if that stops, erases all that chain c has written, restores y, and then does what A does on y. If chain c stops on x, B behaves exactly as A and has P; if not, B never stops on any input and has not P. Applying ⟨Judge-P⟩ to B would thus decide whether chain c stops on x, contrary to Theorem 2.

"Stops on every input", "on some input outputs the command to open a valve", and "behaves exactly as a certain chain already passed by the inspectors" are all properties of this kind. Actions taken in the course of the working — whether a command to open a valve, or to drain the mana core attached, is ever issued along the way — are not properties of behaviour in the sense of Theorem 4, yet they too admit no total judge: make a chain which first carries out the working of chain c on x, issuing neither command meanwhile, and only after it stops issues one; to judge whether it ever does so is to judge whether chain c stops on x. What is spoken of here is whether the chain issues the command; whether the valve in fact opens, or the core is in fact drained, depends on the engine and its supply, and lies outside the model of this paper. There is, therefore, no program which, for every formula whatever that may be sent to it, answers rightly "passed" or "not passed". An inspection may answer "not proved"; it may require a formula to carry its own proof; it may confine formulae to a class that can be completely judged; or it may make honest trial. Where a proof is carried, the proof assures the conclusion; where the class is confined, the judgment assures it; where trial is made, only the cases tried are assured.

Theorem 5. *If a chain may use only a fixed number n of cells — its input being written within the first n, and a move to the right from the n-th cell leaving it where it is — then whether it stops on a given input can be decided.*

Proof. The situation of such a chain is fixed by three things — the contents of the band, the present unit, and the cell under reading — each of which can be in only finitely many states: if the chain has q units besides the stop and the table has k marks, there are at most $N = qnk^n$ situations short of the stop. Let the chain run N steps, passing through $N + 1$ moments counting the first. If it has not stopped, some situation has occurred twice, and from then on it can only repeat the same course for ever.

N grows very fast with n: if the table of marks has k marks, the contents of the band alone can be in k^n states. "Decidable in principle" is therefore not "decidable in practice". In practice three classes are commonly met with that can be completely judged: chains without loops, which always stop; chains in which the number of passes round each loop is bounded beforehand by an amount reckonable from the length of the input, which also always stop; and chains with a fixed number of cells, which can be judged by Theorem 5. Most control formulae of the workshops can be written in the first two classes. Finley's method works because it uses the structure of control formulae rather than enumerating situations one by one. The sixth of the Millennium Seven [6] asks whether problems whose answers can be quickly checked once given can also have their answers quickly found. That is a question of speed; this paper is concerned only with possibility, and does not touch it.

Theorem 6. *Let there be a system all of whose proofs can be listed one by one by a chain, in which propositions of the form "chain c never stops on x" can be recognised mechanically, and all of whose proved propositions of this form are true. Then there is a pair (c, x) such that chain c in fact never stops on x, but the system cannot prove it.*

Proof. If not, then for every pair (c, x) for which chain c never stops, the system proves "chain c never stops on x". Any pair could then be decided thus: run chain c on x, and at the same time list the proofs of the system one by one; if the former stops, answer "stops"; if a proof of "chain c never stops on x" appears, answer "never stops". One of the two must happen after finitely many steps, and since what the system proves is true, the answer is right. This would be a total judge, contrary to Theorem 2.

The Babel fragment released in F.E. 290 [5] proves that a system which does not contradict itself, can express the arithmetic of the natural numbers, and has rules that can be listed, cannot prove by its own means that it does not contradict itself. The fragment starts from within the system, this paper from the one who judges, and the two roads lead to neighbouring places. It should be said that Theorem 6 requires the system's assertions that a chain never stops to be sound, while the fragment requires the system to be free of contradiction and able to express arithmetic; the two sets of conditions are different, and this paper does not derive the fragment's result.

12. On insight

It may be said that the seers of the White Tower sometimes see at a glance how a formula will end, and not by rule. On this the theorems have nothing to say, for they concern judgment by rule. Two things may nevertheless be said.

First, by the rule laid down by the White Tower Society about C.R. 1090, that insight must be reckoned, whatever is gained by insight becomes learning only when it is written in formulae and checked. Checking is of two kinds: the examining of a written proof, which is a matter of rule, and the checking by trial of casting, which can only go case by case. What the theorems of this paper govern, then, is the examination that follows reckoning, not insight itself; how insight arises I do not discuss.

Secondly, if there were a seer who could see, for every rune chain, whether it stops, and were never wrong, his gift, taken as a uniform method, could not be written as rules, for written as rules it would be a total judge. His conclusions might still be proved or checked one by one, so far as proof or trial could reach; but by Theorem 6 his judgments that a chain never stops could not all be proved in one sound system, and the gift itself, that of seeing it for every chain, could by the rule never be reckoned. Whether there is any such seer is not a question that arithmetic can answer.

13. Conclusion

The Fifth Question was put one thousand and eleven years ago, and its answer is negative. This does not hinder the craftsmen in their work. It only shows where their labour is best spent: in confining the formulae to be judged to classes that can be decided, in writing a proof for each formula, or in honest trial.

Earlier drafts of this paper were read by colleagues at White Tower University and the University of Kanaich. The construction of Theorem 2 came from a conversation with an old rune-cutter of the Guild of Rune Mages, who told me that the apprentices' oldest trick for teasing an inspector is to hide in a formula a passage which first reads the mark the inspector has stamped, and then does the opposite.

Appendix. Reckonable and recursively definable functions

The recursively definable functions of Prescott [8] are those obtained from three initial functions — the function that is always nought, the function that adds one, and the function that picks out the i-th argument — by three operations: composition, recursion and least search. Least search means this: given a function g, for each set of arguments try $y = 0, 1, 2, \ldots$ in turn: if g is defined at y and is nought, take this y; if it is defined and not nought, try the next; if a y is met at which g is undefined, or if none is ever found, the function is undefined for those arguments. The recursion meant is primitive recursion, the arguments are natural numbers, and signed integers are coded as natural numbers. I show that a function is recursively definable if and only if it is reckonable, "reckonable" being here widened to include functions undefined at some arguments, at which the chain does not stop. Restricted to functions defined everywhere, the two definitions again describe the same class.

(i) Every recursively definable function is reckonable. The proof is by induction on the number of steps in the definition. The three initial functions are evidently reckonable. If g and h are reckonable (each chain first copying the arguments and the step count to a safe place on the band, lest its working erase them), so is their composition, by concatenating the two chains; so is a function defined from them by recursion, by a loop that works up from nought to the place required; and so is least search on g, by a loop that tries one value after another, stopping when g is nought and going on for ever if it never is, which answers exactly to being undefined.

(ii) Every reckonable function is recursively definable. Code the situation of a chain at a given moment — the contents of the band, the present unit and the cell under reading — as a number, in the manner of §5. Division, remainder, the taking out of a stretch of figures, the joining of two stretches and the looking up of an entry in a finite table are all recursively definable, as Prescott [8] has shown; looking up the table, rewriting one cell and moving one cell are compounded of them, so that the transformation by which one step of the chain carries one situation into the next is recursively definable. If a situation already in the stop is taken to pass by one step into itself, the transformation is defined for every situation. The situation after t steps from the input is defined by recursion on t; the number of steps needed to stop is found by least search, as the first t at which the chain is in the stop; and the output is extracted from the situation at stopping.

The two definitions therefore describe the same functions, and the second reason of §9 receives an exact illustration.

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